Mathematics without biology or biology without mathematics are not as wonderful.

My *NeuroAI Research Group* will initially focus on creating advanced statistical methods to characterise neural and behavioural data, offline and in real time, for short- and long-duration experiments. In addition, in collaboration with experimental groups, we will apply these methods, or those created by others, to investigate behavioural and neural questions.

This is an exciting time to work on NeuroAI. As the variety of neural datasets being recorded (e.g., high-channel-count electrophysiological, one- and two-photon calcium imaging, voltage imaging, EM, transcriptonics, connectomics) keeps increasing, and their size and complexity become larger and larger, due to more naturalistic and longer experimentation, there is a critical need for statistical methods that can draw efficient inferences and extract meaning from these datasets. In addition, as new optogenetic methods are developed to spatio-temporally manipulate neural activity, we need these inferences in real time, combined with methods from optimal control, to maximise the efficiency of these manipulations. Yet, we are currently not seeing an increase in the number of NeuroAI scientists, trained in both statistical machine learning and Neurosciences, required to optimally process these datasets with advanced statistical ML methods. My research group will attempt to reverse this trend.

Latent variable models Extracting meaningful structure from the massive spike trains generated by high-channel-count probes like Neuropixels is notoriously challenging. To address this, Gaussian Process Factor Analysis (GPFA; Yu et al., 2009) was developed to extract low-dimensional latent variables from binned spike counts. This approach was recently extended into Sparse Variational GPFA (svGPFA; Duncker and Sahani, 2018) to bypass standard scalability limits. By utilising sparse-variational approximations, svGPFA substantially reduces computational and memory bottlenecks, allowing it to successfully characterise much longer experimental trials than standard GPFA. Furthermore, its continuous-time point-process observation model eliminates the need of binning spikes, and yields a fully continuous neural population model.

At the Gatsby Unit, I developed and distributed a highly optimised Python implementation of svGPFA. To demonstrate the power of this method on complex, large-scale data, I used it to characterise several high-channel count neural datasets. In collaboration with the Stephenson-Jones laboratory at the SWC, we identified a powerful validation of this framework. I applied svGPFA to striatal Neuropixels recordings from mice executing a stereotyped poking sequence task (Thompson et al., 2024). By feeding these neural latents into a Hidden Markov Model (HMM), I decoded distinct behavioural states with high fidelity. Crucially, we have now observed that these latent neural patterns actively replay during sleep. These population replay observations are robust and are not artefacts of svGPFA, as the same replay dynamics are consistently recovered when estimating latents using linear dynamical systems fitted to raw spike counts rather than point-process observations. These findings demonstrate that accurate decoding and replay tracking can be achieved directly within the continuous latent space of large neural populations, completely bypassing the traditional requirement to estimate explicit place fields.

My independent research group will continue developing advanced latent variable models for population recordings. We will collaborate closely with experimental labs at NPP, CDB, and elsewhere to characterise novel datasets and map fundamental brain functions. An immediate focus of my group will be applying latent variable models to investigate position decoding and replay dynamics within hippocampal circuits.

NeuroAI methods for naturalistic, long-duration and continual experimentation At the SWC and Gatsby Unit we are pioneering a new type of naturalistic, long-duration and continual experimentation, implemented in the Aeon platform, which is poised to revolutionise neuroscience experimentation (Campagner et al., 2025).

In 2023–2024, I mentored an exceptional undergraduate student from NPP, Ms. Zimo Li, through a Simons Foundation SURFiN fellowship. Zimo performed the first comprehensive statistical analysis of Aeon data, an outstanding achievement given the high complexity and scale of the unstructured raw datasets at that stage. She skilfully applied sophisticated statistical frameworks—including linear dynamical systems, hidden Markov models, and point-process regression—to generate the first statistical discoveries from the Aeon platform. Her contributions earned her a co-authorship on the Aeon platform paper. This highly successful collaboration demonstrated my capacity to effectively guide life-sciences students through advanced mathematical workflows, enabling them to produce rigorous, high-impact computational research.

Due to the large size of Aeon recordings, conventional batch machine learning (ML) methods cannot be used to process Aeon data. Aeon requires online methods that are able to process non-stationary data and are computationally very efficient to handle very large datasets. A detailed description of my ML data processing plans for Aeon data appears in a collaborative BBSRC/NSF grant we submitted jointly with the Allen Institute for Neural Dynamics in 2024. My group will continue contributing advanced statistical methods for the characterisation of Aeon data offline and in real time.

NeuroAI methods for intelligent experimental control Current neuroscience experimentation is mostly performed in open loop with predefined sets of rules designed by experimenters. More informative experiments can be designed if they could be controlled in close loop by intelligent inferences performed on behavioural and neural data streams.

Bonsai is today the most widely used software for close loop experimental control in neuroscience. In 2022 I realised that adding ML functionality to Bonsai could make possible a radically new type of close loop intelligent experimentation, and led the successful application of a BBSRC proposal (BB/W019132/1) to build the Bonsai.ML package.

Most current ML methods operate offline, trained and performing inference on datasets previously recorded. However, Bonsai.ML requires a different type of ML methods, which are trained and perform inference on live streams, continuously delivering data in real time. We built this type of ML into Bonsai.ML. It now contains probabilistic ML methods for real time processing of behavioural (e.g., kinematics inference with linear dynamical systems, behavioural state estimation with hidden Markov models) and neural data (neural latent variables inference with linear dynamical systems, hippocampal position decoding and replay detection with a point process state-space model). Please refer to the documentation of Bonsai.ML.

We are now integrating large language models (LLMs) into Bonsai. We are using them to help Bonsai users build correct and optimised workflows using natural language. Additionally, we are building a Bonsai LLM node to bring the large semantic knowledge of LLMs into the experimental control loop. Please refer to the Open Source for the Life Science proposal we are currently preparing to support these developments.

My group will continue developing real-time probabilistic ML models to enable intelligent experimental control in Bonsai.

Collaborations In 2019, I was hired by the Gatsby Unit specifically to facilitate interactions between the Unit and the SWC, by collaborating with SWC scientists on the use of advanced statistical methods. Below I describe the dynamics of one such collaboration.

To identify collaborative opportunities, I regularly participated in the SWC Data Clubs. At one session, Dr. Mitra Javadzadeh (then a PhD student in Prof. Hofer's lab) presented her work on long-range cortical communication using Linear Discriminant Analysis. I proposed that a linear dynamical systems framework would better capture the intrinsic temporal characteristics of her experimental setup. This led to a close collaboration where we applied these models to her data, yielding compelling insights that Dr. Javadzadeh later presented as an oral contribution at CoSyNe 2022.[1] Now a faculty member at Cold Spring Harbor Laboratory (CSHL), Dr. Javadzadeh continues to build upon this specific research line (Javadzadeh et al., 2024).

This partnership, alongside my work with other SWC investigators, underscores how deeply methodological-experimental collaborations shape both scientific trajectories and career paths. Driven by these insights, I am currently writing an article for The Transmitter on the critical need for more NeuroAI scientists within the broader neuroscience community. To extract maximum meaning from modern, high-density neural recordings, we need researchers trained rigorously at the intersection of statistical machine learning and systems neuroscience. As a faculty member, my goal is to establish a dedicated team of NeuroAI researchers who will deeply integrate with experimental laboratories to develop optimal, bespoke modelling frameworks for complex datasets.

Teaching Vision My teaching of quantitative subjects to life-sciences students balances rigorous mathematical foundations with practical applications. This philosophy was shaped during my time at Brown University under the mentorship of Prof. Wilson Truccolo, for whom I served as a teaching assistant in his graduate-level Statistical Neuroscience course. Prof. Truccolo's lectures masterfully combined foundational mathematical concepts with fascinating neuroscience applications, frequently drawing from his own research. I have integrated this balance into my own instructional strategy, ensuring that mathematical concepts are anchored by biological relevance.

I structure my lectures to be multidimensional, targeting both novices and advanced students. I maintain a streamlined narrative during class by moving mathematical proofs out of the main thread. However, I provide these proofs in appendices. My lectures contain multiple examples, usually in the form of computer exercises. By providing boilerplate code, I help students perform advanced neural data analysis early on. This approach ensures that students experience the reward of mastering complex processing, even as they are still digesting the underlying mathematical concepts.

I have successfully applied this methodology across several courses. For example, I designed the Probability module of the inaugural Gatsby Unit Bridging Program in 2023. There I delivered the Foundations of Probability Theory lecture, which serves as a strong benchmark of my teaching style. Additionally, for first-year PhD students at the Sainsbury Wellcome Centre (SWC), I designed practical assignments connecting statistical concepts with neuroscience applications. In a homework assignment on multivariate linear regression, students utilised my boilerplate code to fit nonlinear visual receptive fields models to spiking activity from cat primary visual cortex (Rapela et al., 2006). Similarly, in a homework assignment on circular statistics, I guided them to reproduce circular statistics and travelling wave detection figures from Rapela (2016, 2017, 2018).

I am passionate about the subjects I teach, and from student feedback I learned that this passion is reflected in my teaching.

[1] Mitra presented orally our results at CoSyNe 2022. Presentation begins at 2:59:00.

References

Campagner, D., Bhagat, J., Lopes, G., Calcaterra, L., Pouget, A., Almeida, A., Nguyen, T., Lo, C., Ryan, T., Cruz, B., et al. (2025). Aeon: an open-source platform to study the neural basis of ethological behaviours over naturalistic timescales. *bioRxiv*, pages 2025–07.

Duncker, L. and Sahani, M. (2018). Temporal alignment and latent gaussian process factor inference in population spike trains. In *Advances in Neural Information Processing Systems*, pages 10445–10455.

Javadzadeh, M., Schimel, M., Hofer, S. B., Ahmadian, Y., and Hennequin, G. (2024). Dynamic consensus-building between neocortical areas via long-range connections. *bioRxiv*.

Rapela, J. (2016). Entrainment of traveling waves to rhythmic motor acts.

Rapela, J. (2017). Rhythmic production of consonant-vowel syllables synchronizes traveling waves in speech-processing brain regions.

Rapela, J. (2018). Traveling waves appear and disappear in unison with produced speech.

Rapela, J., Mendel, J., and Grzywacz, N. (2006). Estimating nonlinear receptive fields from natural images. *Journal of Vision*, 6(4):441–474.

Thompson, E., Rollik, L., Waked, B., Mills, G., Kaur, J., Geva, B., Carrasco-Davis, R., George, T., Domine, C., Dorrell, W., et al. (2024). Replay of procedural experience is independent of the hippocampus. *bioRxiv*, pages 2024–06.

Yu, B. M., Cunningham, J. P., Santhanam, G., Ryu, S. I., Shenoy, K. V., and Sahani, M. (2009). Gaussian-process factor analysis for low-dimensional single-trial analysis of neural population activity. *Journal of neurophysiology*, 102(1):614–635.